

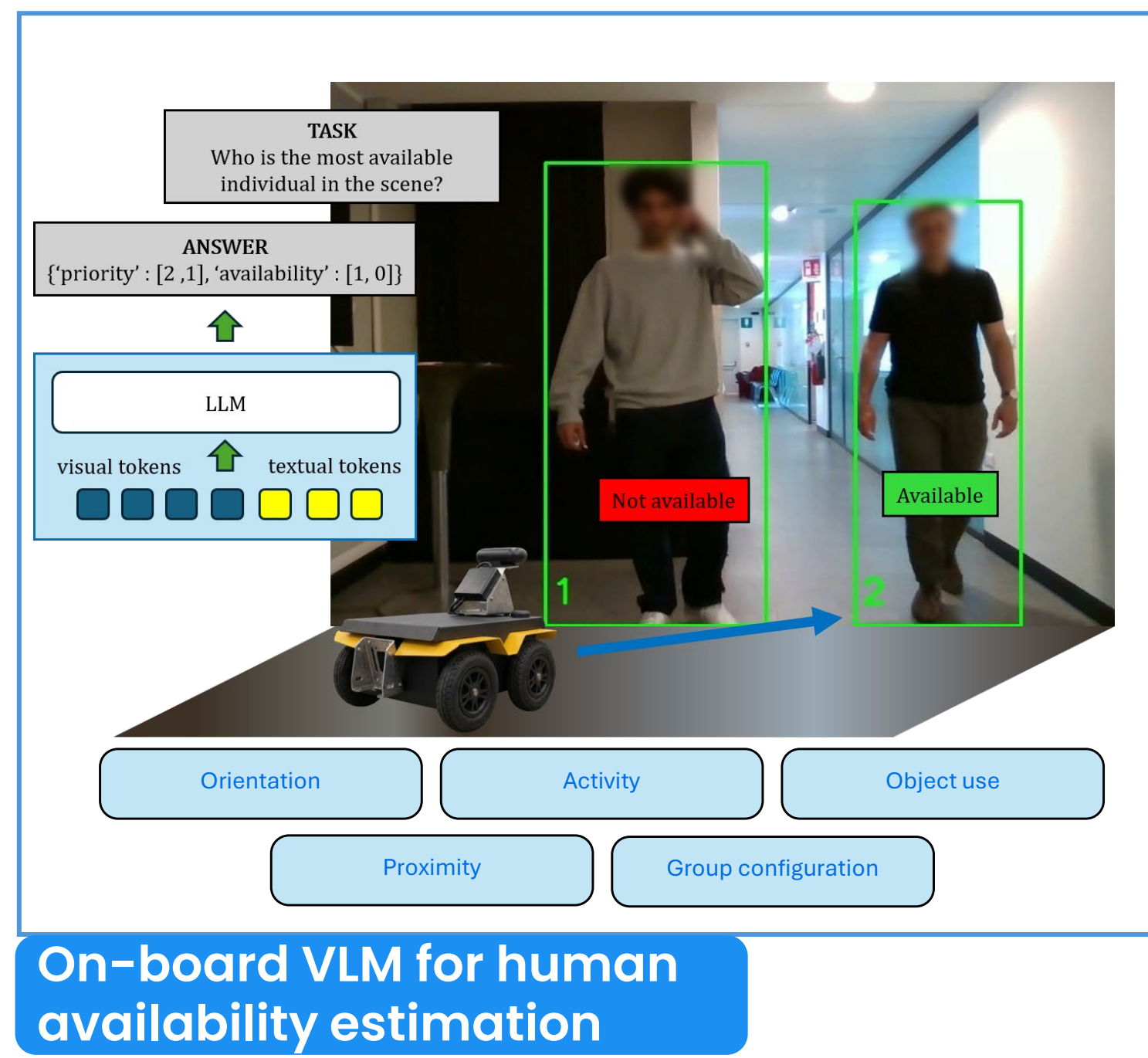
Enabling Effective Human-Robot Collaboration through Efficient On-Board Vision-Language Models

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1. MOTIVATION

- Collaborative robots** must interpret **socially relevant cues**, going beyond geometric perception to support socially appropriate interactions.
- Before initiating an interaction, a robot must determine whether a person is **available to be approached**, rather than only estimating engagement once an interaction has already started.
- Multi-person environments** make this decision more challenging, as several potential interaction partners must be evaluated simultaneously and prioritized.
- Large VLMs provide strong semantic understanding but are difficult to **deploy on-board**, due to computational cost, connectivity dependence and privacy concerns.



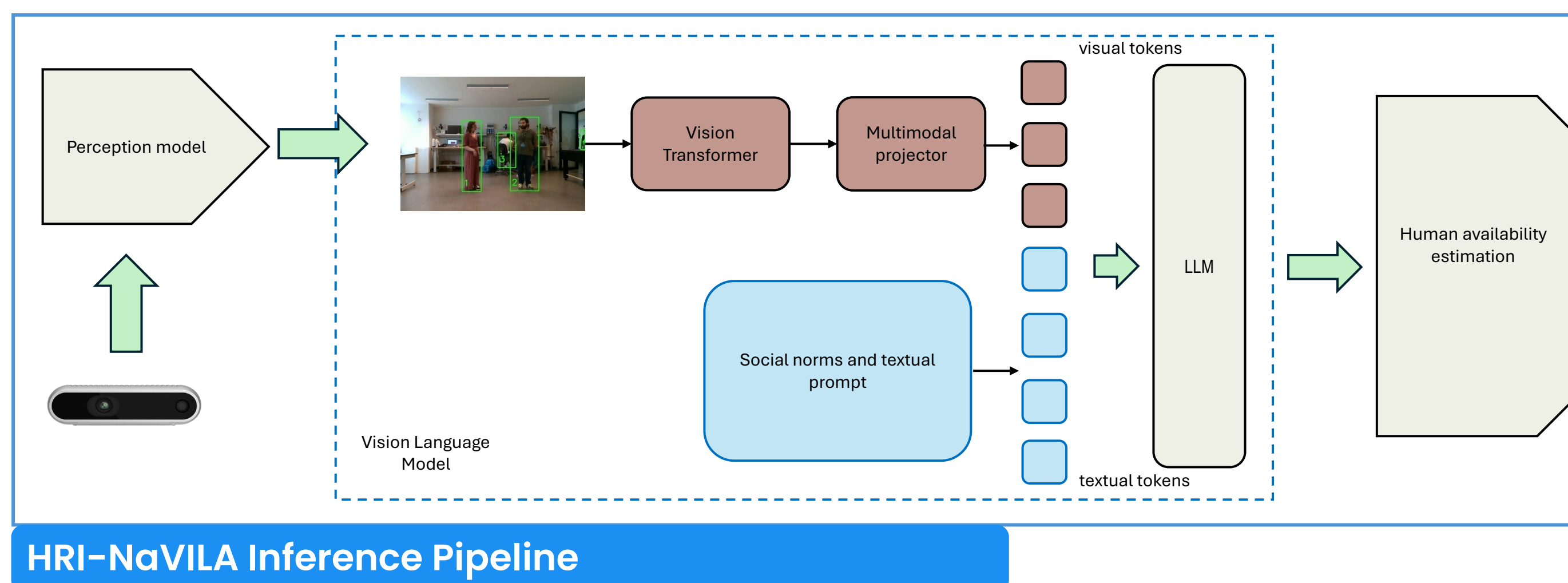
2. DATASET CREATION

- Custom **indoor HRI dataset** targeting human interaction availability.
- Training recordings vary **posture, orientation, activity, interpersonal distance and social configuration**.
- Frames sampled at **0.5–1 fps** to reduce temporal redundancy.
- Validation and test sets include more naturalistic interactions, **unseen participants** and varying **lighting conditions**.



3. METHODOLOGY

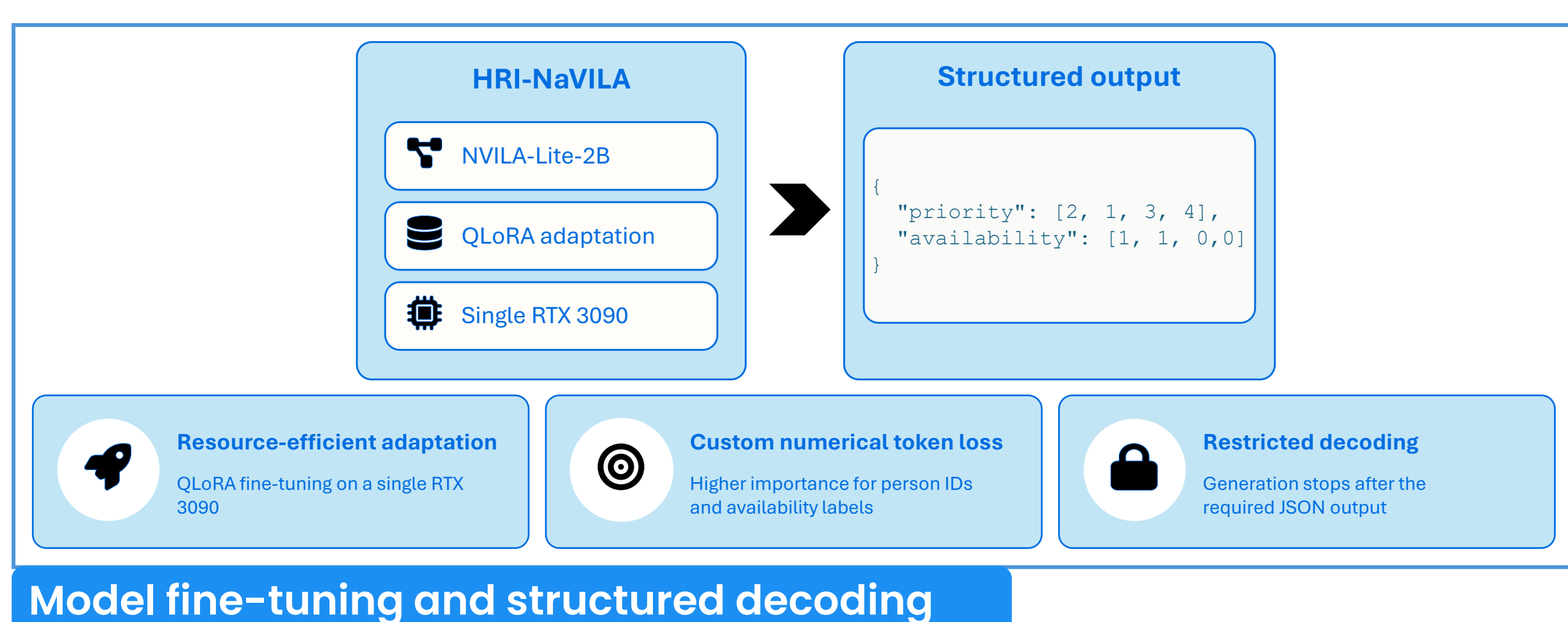
- Person grounding**: a perception model detects people in the RGB observation and associates each individual with a **numerical ID**.
- Multimodal social reasoning**: HRI-NaVILA combines visual features with a textual prompt encoding the task and social norms to estimate interaction availability in multi-person scenes.



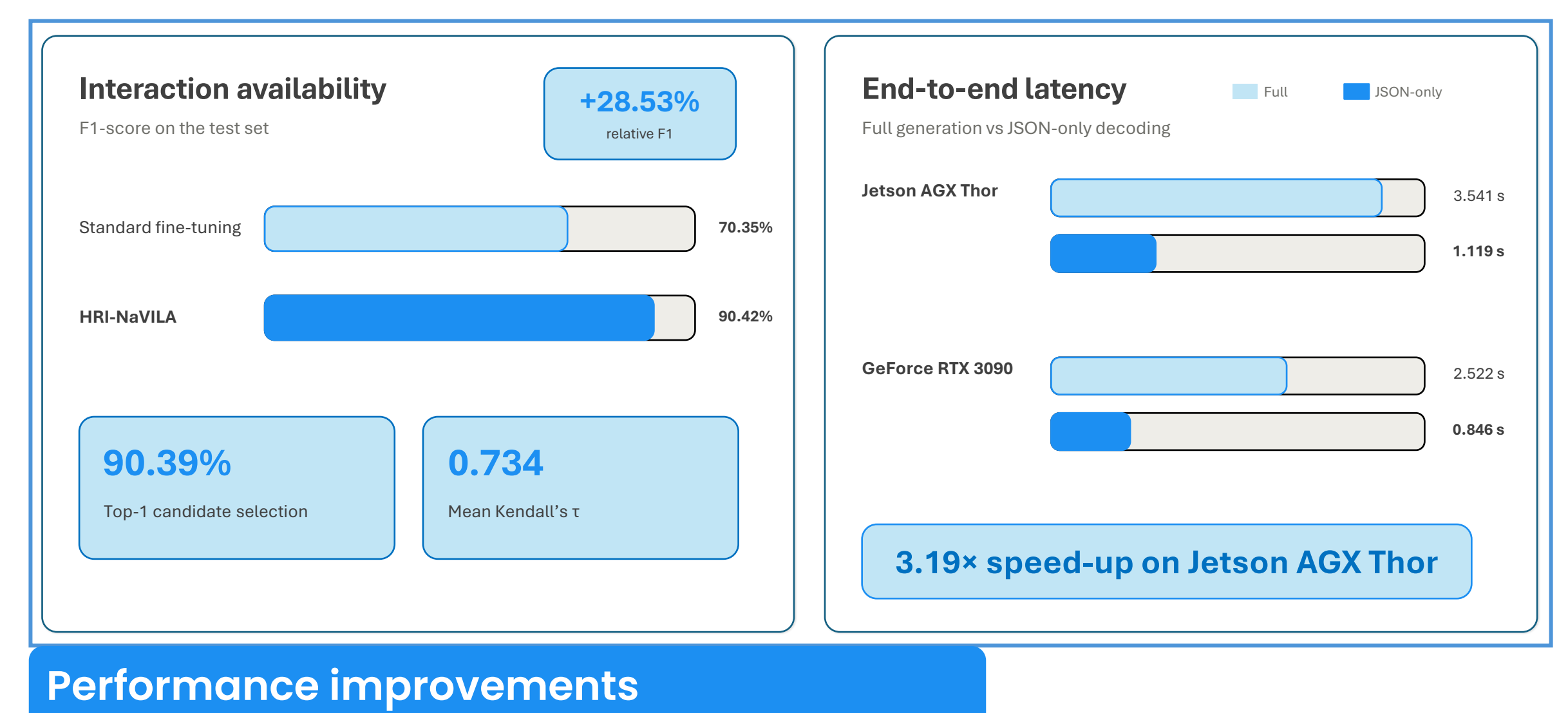
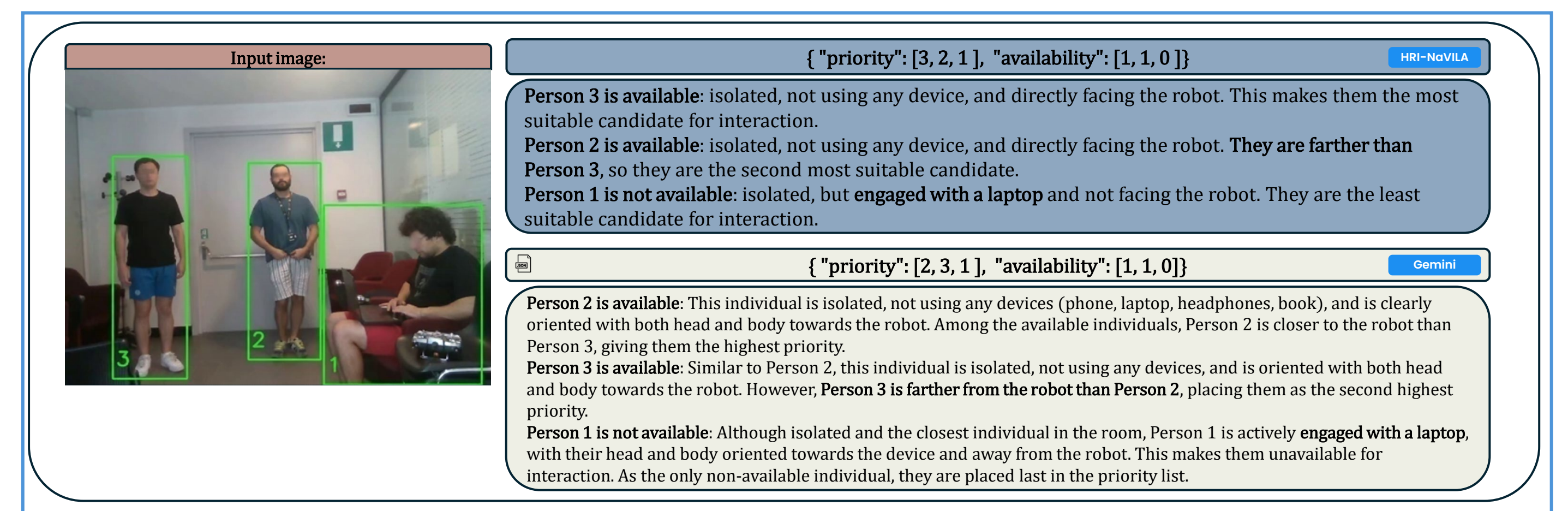
- Resource-efficient fine-tuning**: QLoRA adaptation of **NVILA-Lite-2B¹**, performed on a single **NVIDIA GeForce RTX 3090 GPU**.
- Task-relevant token weighting**: Numerical tokens encoding **person IDs** and **availability labels** receive a larger contribution during fine-tuning because errors on these values directly affect the robot's decision.

$$\mathcal{L}_{TWCE} = - \sum_{w=1}^{|S|} \sum_{e=1}^{|V|} (1 + (\lambda - 1) \mathbf{1}_{\text{num}}(e)) \cdot y_{w,e} \log(\hat{y}_{w,e})$$

- Structured decision output**: the model directly predicts a **priority ranking** and a **binary availability label** for each detected person, providing the information required for downstream robot decision-making.



4. RESULTS



- HRI-NaVILA achieves performance close to **Gemini-2.5-Flash²** and **GPT-5-mini³** in few-shot evaluation, while enabling fully on-board inference with a lightweight model.
- Compared with **FlowAct IAB⁴**, contextual VLM reasoning substantially improves availability estimation when interaction readiness depends on activities, objects and social configurations beyond **gaze** and **head pose**.

Models	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	$\bar{r}$	Top1-acc.	FIR(%)
Gemini-2.5-Flash	97.59	95.89	96.59	97.47	96.24	0.7342	91.42	4.01
GPT-5-mini	95.75	90.92	98.28	94.07	94.46	0.8491	95.88	10.58
FlowAct IAB	65.74	83.01	10.65	98.66	18.88	0.5125	79.75	1.82
HRI-NaVILA-base	82.28	91.08	57.31	96.59	70.35	0.6150	87.47	4.00
HRI-NaVILA	94.13	93.24	87.77	96.19	90.42	0.7342	90.39	4.38

Interaction condition	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	$\bar{r}$	Top1-acc. (%)	FIR (%)
Single person	95.35	81.66	84.48	91.91	83.05	-	100.00	0
One available	98.62	99.07	98.16	99.08	98.62	1.0000	100.00	0
Multiple available	84.44	100.00	82.14	100.00	90.19	0.6122	75.51	0
Group interaction	96.56	89.36	93.33	97.34	91.30	0.5823	80.45	6.97
No available	95.94	-	-	95.94	-	0.6981	84.90	8.49

References

- Z. Liu, L. Zhu, B. Shi, Z. Zhang, and L. et al., "Nvila: Efficient frontier visual language models," in 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
- Comanici, E. Bieber, M. Schaeckermann, and I. P. et al., "Gemini2.5: Pushing the frontier with advanced reasoning, multimodality, longcontext, and next generation agentic capabilities," 2025.
- OpenAI, "Gpt-5 system card," 2025, available: <https://cdn.openai.com/gpt-5-system-card.pdf>.
- T. Dhaussy, B. Jabaian, and F. Lefevre, "Flowact: A proactive multimodal human-robot interaction system with continuous flow of perception and modular action sub-systems," in Proceedings of the 14th International Conference on Pattern Recognition Applications and Methods, ICPRAM 2025.

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